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Detection and characterisation of a sixth *Candida auris* clade in Singapore: a genomic and phenotypic study



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Summary

Background The emerging fungal pathogen *Candida auris* poses a serious threat to global public health due to its worldwide distribution, multidrug resistance, high transmissibility, propensity to cause outbreaks, and high mortality. We aimed to characterise three unusual *C. auris* isolates detected in Singapore, and to determine whether they constitute a novel clade distinct from all previously known *C. auris* clades (I–V).

Methods In this genotypic and phenotypic study, we characterised three *C. auris* clinical isolates, which were cultured from epidemiologically unlinked inpatients at a large tertiary hospital in Singapore. The index isolate was detected in April, 2023. We performed whole-genome sequencing (WGS) and obtained hybrid assemblies of these *C. auris* isolates. The complete genomes were compared with representative genomes of all known *C. auris* clades. To provide a global context, 3651 international WGS data from the National Center for Biotechnology Information (NCBI) database were included in a high-resolution single nucleotide polymorphism (SNP) analysis. Antifungal susceptibility testing was done and antifungal resistance genes, mating-type locus, and chromosomal rearrangements were characterised from the WGS data of the three investigated isolates. We further implemented Bayesian logistic regression models to classify isolates into known clades and simulate the automatic detection of isolates belonging to novel clades as their WGS data became available.

Findings The three investigated isolates were separated by at least 37 000 SNPs (range 37 000–236 900) from all existing *C. auris* clades. These isolates had opposite mating-type allele and different chromosomal rearrangements when compared with their closest clade IV relatives. The isolates were susceptible to all tested antifungals. Therefore, we propose that these isolates represent a new clade of *C. auris*, clade VI. Furthermore, an independent WGS dataset from Bangladesh, accessed via the NCBI Sequence Read Archive, was found to belong to this new clade. As a proof-of-concept, our Bayesian logistic regression model was able to flag these outlier genomes as a potential new clade.

Interpretation The discovery of a new *C. auris* clade in Singapore and Bangladesh in the Indomalayan zone, showing a close relationship to clade IV members most commonly found in South America, highlights the unknown genetic diversity and origin of *C. auris*, particularly in under-resourced regions. Active surveillance in clinical settings, along with effective sequencing strategies and downstream analysis, will be essential in the identification of novel strains, tracking of transmission, and containment of adverse clinical effects of *C. auris* infections.

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Introduction

Candida auris is an emerging multidrug-resistant (MDR) fungal pathogen that has been implicated in many infection outbreaks associated with health care.^{1–4} Invasive infections caused by MDR *C. auris* are associated with high mortality in vulnerable patient groups.^{1,2,4–6} Within health-care settings, *C. auris* appears to be highly transmissible and is known to persist in the environment.^{3,4,7,8} Given its worldwide distribution, MDR characteristics, high transmissibility, and high mortality, *C. auris* represents a serious threat to global public health. WHO highlighted this threat by classifying *C. auris* as

a critical priority pathogen,⁹ and the US Centers for Disease Control and Prevention declared *C. auris* as an urgent antimicrobial resistance threat,¹⁰ further emphasising the need to understand and mitigate this public health threat.

Five geographically distinct *C. auris* clades have been described: clade I (south Asian), clade II (east Asian), clade III (African), clade IV (South American), and clade V (Iranian).^{11,12} These clades were recognised based on analyses of whole-genome sequencing (WGS) data and single nucleotide polymorphism (SNP) variation, where strains from different clades differed in tens of thousands

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Research in context

Evidence before this study

We searched PubMed using the search terms “*Candida auris*” AND “clade” for papers published between Jan 1, 2009, and July 1, 2023, with no language restrictions. This search retrieved 115 publications. 60 relevant publications were identified. 28 studies analysed and discussed the molecular epidemiology of *C auris*, including the description of *C auris* clades, either in outbreak or surveillance settings. There were 11 case reports of *C auris* clinical cases that included clade determination. Two studies focused on the detection and clade determination of *C auris* from non-health-care environments. Clade-specific characteristics were described or analysed in 14 studies. One study applied machine learning to *C auris* drug resistance analysis, but not for clade determination. Four studies focused on the description of potentially new *C auris* lineages, subclades, or clades. All publications described isolates that belong to one of the five known *C auris* clades (I–V). All publications found that strains from different clades differed by at least tens of thousands of single nucleotide polymorphisms (SNPs), and that there are clade-specific differences in geographical distribution, phenotypic characteristics, antifungal susceptibility profile, outbreak potential, and clinical manifestations. The National Center for Biotechnology Information (NCBI) Pathogen Detection system contained 4506 *C auris* genomes on July 1, 2023. There were ten (0.22%) submissions from southeast Asian countries and 92 (2.04%) submissions from south Asia and the Indian subcontinent.

Added value of this study

To our knowledge, we are the first group to perform hybrid assemblies on three representative isolates in a new *C auris* clade, which is separated from all other existing clades (I–V) by at least

37 000 SNPs (range 37 000–236 900). Whole-genome SNP analysis and phenotypic characterisation of these epidemiologically unlinked isolates detected in Singapore suggest that they represent a previously unreported sixth major clade. High-resolution SNP analysis of 3651 international whole-genome sequences from the NCBI database, which generated the final dataset consisting of more than 6.6 million genome pairs, revealed six distinct genetic clusters representing the five known clades and the new sixth clade. In addition, we showed that a machine learning approach can potentially be used to automatically classify genomes and flag outlier genomes for further investigations as soon as they become available, thus providing the earliest possible alert for potential new public health threats.

Implications of all the available evidence

Despite the high antimicrobial resistance burden in southeast Asia and south Asia, these regions are disproportionately under-represented in terms of genomic surveillance of *C auris*, a multidrug-resistant fungal pathogen. The detection of three epidemiologically unlinked *C auris* isolates in Singapore belonging to a new *C auris* clade suggests that yet-to-be-reported strains might be circulating in the region. Given the propensity for multidrug resistance, health-care-associated infection outbreaks, and the associated high mortality, active surveillance and continued vigilance are necessary. The development and use of machine learning approaches to automatically detect potentially novel genomes and derive actionable insights as soon as the genome data become available could enhance our ability to detect emergent public health threats.

of SNPs.^{11–13} Furthermore, these clades differed in terms of their antifungal susceptibility profiles, outbreak potential, and clinical manifestations.¹⁴

We aimed to characterise three epidemiologically unlinked *C auris* isolates detected in Singapore, which had unusual phenotypic characteristics. The index isolate (isolate A, LIMS ID F3485) was first detected by PCR from a routine *C auris* screening swab and subsequently cultured in April, 2023. The screening swab was taken within 24 h of patient A's admission as part of a *C auris* active surveillance programme in the hospital. Patient A received treatment in the Department of Internal Medicine at Singapore General Hospital (SGH) and reported no travel history in the preceding 2 years. Because *C auris* was not known to be endemic in Singapore, epidemiological, microbiological, and molecular investigations were initiated. Isolate A was initially suspected to belong to clade II in view of the available epidemiological data and susceptibility to antifungals. However, unlike clade II isolates, isolate A grew at 42°C and was positive for L-Rhamnose and N-acetyl-glucosamine assimilation. Retrospective analysis of our laboratory's archived *C auris*

isolates found two additional isolates (B and C) with similarly unusual phenotypic characteristics.

Isolate B (LIMS ID F1580) was cultured from an intra-operative tissue specimen from patient B in April, 2022. Patient B received treatment in the orthopaedic unit at SGH and reported no travel history in the preceding 2 months; however, the patient did have outpatient visits at hospitals in Indonesia in 2021. Isolate C (LIMS ID F0083) was cultured from the positive blood culture of patient C in January, 2018. Patient C was airlifted from Bangladesh to the burns unit at SGH, and the blood culture taken within 48 h of admission grew *C auris*. Isolate C was previously sequenced in 2019 and the data are publicly available (SRR10102336).¹⁵ Given the incongruent epidemiological data, antifungal susceptibility pattern, and biochemical profiles of these three isolates, we hypothesised that they might be representatives of a new clade.

We aimed to characterise these three unusual *C auris* isolates detected in Singapore and to establish whether they constitute a novel clade distinct from all previously known *C auris* clades (I–V). We further aimed to simulate the automatic detection of the new clade, to assess whether

	Isolate A	Isolate B	Isolate C
Patient identifier	A	B	C
Laboratory information management system identifier	F3485	F1580	F0083
Source	Nasal, axillae, and groin swab	Intraoperative tissue	Blood
Date of clinical sample collection	April 19, 2023	July 4, 2022	Jan 3, 2018
Existing clade identified	None	None	None
First sequenced in this study	Yes	Yes	No
Accession before this study	Not available	Not available	BioProject PRJNA540907, BioSample SAMN11866133
Accession generated in this study	BioProject PRJNA1000034, Biosample SAMN36753178	BioProject PRJNA1000034, Biosample SAMN36753179	BioProject PRJNA1000034, Biosample SAMN36753180

Table 1: Metadata of the three investigated Singapore *Candida auris* isolates

microbial genomic surveillance can be augmented to provide early alerts for potential emergent threats.

Methods

Study design, *C. auris* isolates, and epidemiological data

Three *C. auris* isolates (A, B, and C) identified at Singapore General Hospital were included in this phenotypic and genomic analysis. Four American Type Culture Collection (ATCC; Manassas, VA, USA) *C. auris* isolates from clades I–IV were included as clade-specific controls in this study (appendix 1 p 1). We used pre-existing retrospective collections of isolates, and epidemiological data (appendix 2 p 34) were previously collected as part of routine surveillance and infection prevention measures (appendix 2 pp 4, 23). Institutional review board exemption was granted by the SingHealth Centralised Institutional Review Board (reference number 2017-2576).

Procedures

All isolates in the study were identified using matrix-assisted laser desorption ionisation time-of-flight mass spectrometry (Bruker Daltonik, Leipzig, Germany) and sequencing of internal transcribed spacer region. Phenotypic biochemical profiles were obtained using the Vitek 2 YST ID Card (bioMérieux, Marcy-l'Étoile, France) and API 20C AUX test kit (bioMérieux, Marcy-l'Étoile, France). Antifungal susceptibility testing was performed using the Sensititre YeastOne ITAMYUCC plate for nine antifungals and YO10 AST plate for flucytosine; (Sensititre TREK Diagnostic Systems, East Grinstead, UK) according to the manufacturer's instructions in a College of American Pathologists-accredited clinical laboratory at SGH. The aggregation assay was performed as previously described¹⁶ (appendix 2 p 4).

We sequenced the whole genomes of seven *C. auris* isolates, including the three investigated isolates A, B and C, and four ATCC control isolates for clades I–IV (appendix 1 p 1). All seven isolates were long-read sequenced using the MinION sequencing platform (Oxford Nanopore Technologies, Oxford, UK). Additional short-read Illumina (San Diego, CA, USA) sequencing was performed for the three investigated isolates, and clades I (ATCC5000) and

IV (ATCC5003) ATCC isolates. Hybrid assemblies were obtained for the investigated isolates (A, B, and C), and clade I and clade IV ATCC isolates (appendix 1 p 1). The methods and settings for sample processing, sequence data processing, read quality control, and genome assembly are in appendix 2 (pp 4–6).

For between-clade comparisons, nine additional, publicly available assembled *C. auris* genomes (clades I–IV) were retrieved from the National Center for Biotechnology Information (NCBI) Genome Database and the Candida Genome Database (appendix 1 p 1). Nanopore long-read data of a clade V isolate (B18474, SRR13269545) were downloaded from NCBI's Sequence Read Archive and assembled. These 17 assembled genomes constitute genome representatives of all existing clades (I–V), and the proposed new clade. Core genome SNPs of these assembled genomes were determined with Parsnp, using B8441 as a reference¹⁷ (appendix 2 p 7). To verify that the SNP distances observed between clades were not method-specific, we additionally performed variant calling using a GATK-based pipeline as described in a recent study¹⁸ (appendix 2 p 7). The detailed methods and settings used for variant calling, genetic relatedness, and phylogenetic analysis are in appendix 2 (pp 6–7).

The assembled genomes were characterised for mating-type alleles and the following antifungal resistance

See Online for appendix 1

See Online for appendix 2

For Parsnp see <https://github.com/marbl/parsnp>

For GATK see <https://gatk.broadinstitute.org/hc/en-us>

	MIC of isolate A (F3485), mg/L	MIC of isolate B (F1580), mg/L	MIC of isolate C (F0083), mg/L
Anidulafungin	0-12	0-12-0-25	0-06-0-12
Micafungin	0-12-0-25	0-06-0-12	0-06-0-12
Caspofungin	0-06-0-5	0-03-0-25	0-06-0-12
Flucytosine	≤0-06	≤0-06	≤0-06
Isavuconazole	0-03-0-06	≤0-008-0-03	0-06-0-12
Posaconazole	0-03-0-06	≤0-008-0-015	0-03-0-06
Voriconazole	0-03-0-06	≤0-008-0-015	0-06-0-12
Itraconazole	0-06-0-12	0-03-0-12	0-12-0-25
Fluconazole	2-4	1-2	4-8
Amphotericin B	0-5-1	0-5-1	2

Susceptibility testing was performed in triplicate. The results of the control and reference strains are in appendix 1 (p 20). MIC=minimum inhibitory concentration.

Table 2: MIC values of investigated *Candida auris* isolates

For NCBI Pathogen Detection
see <https://www.ncbi.nlm.nih.gov/pathogens>

For Snippy see <https://github.com/tseemann/snippy>

For BEAST see <https://beast.community/index.html>

For ADMIXTURE see
<https://dalexander.github.io/admixture/>

genes: *ERG11* (encoding lanosterol 14 α -demethylase), *ERG6* (encoding sterol 24-C-methyltransferase), *FKS1* (encoding 1,3-beta-glucan synthase), *TAC1B* (encoding zinc-cluster transcription factor), and *FUR1* (encoding uracil phosphoribosyl-transferase; appendix 2 p 7). Since previous studies observed clade-specific chromosomal rearrangements in *C. auris*,^{19,20} we analysed chromosomal rearrangements by aligning assembled genomes to the reference clade I genome CA-AM1 (GCA_014673535.1). Only contigs and rearrangement events of size greater than 10 000 bp were visualised (appendix 2 p 8).

In silico short tandem repeat (STR) analysis was performed to find out whether the investigated isolates have a distinct STR profile (appendix 2 p 8).²¹ To estimate the divergence time, Bayesian phylogenies were generated using BEAST version 1.10.4 as previously described,²² using clade I *C. auris* genome B11205 (GCA_016772135.1) as a reference. Details of this phylogenetic timing analysis and the list of genomes included in this analysis are available in appendix 2 (pp 8–9).

ADMIXTURE was used to further examine if the investigated genomes are recombinants or hybrids of the existing clades.²³ Cross-validation error analysis was performed across all k values (k values 2–10). The k value with the lowest error was selected for ADMIXTURE analysis with

five replicates with different random number seeds (appendix 2 p 9). 89 *C. auris* genomes from all known clades and the proposed new clade were included in this analysis (appendix 1 p 3).

To do a large-scale analysis of global *C. auris* genomes, a list of 4358 Illumina WGS accessions together with data release date was obtained from NCBI Pathogen Detection on June 29, 2023. This list was then combined with the curated list of 1285 *C. auris* WGS data with clade information reported in previous studies.^{11,24} In total, 4475 unique Illumina WGS data were included in this large-scale analysis, of which 3651 passed quality control and were included in downstream analyses (appendix 1 p 2). Variant calling was performed using Snippy version 4.6.0 with B8441 as the reference genome (appendix 2 p 7). The SNPs identified were used for genome relatedness analysis.

Statistical analysis and machine learning models

Current methods of clade determination rely on phylogenetic analysis or STR genotyping, the interpretation of which can be challenging, leading to uncertainty in clade assignments. There is currently no established SNP distance threshold to flag an unusual *C. auris* genome as a potential new clade. To address this, we hypothesised that a Bayesian logistic regression model could learn clade-differentiating

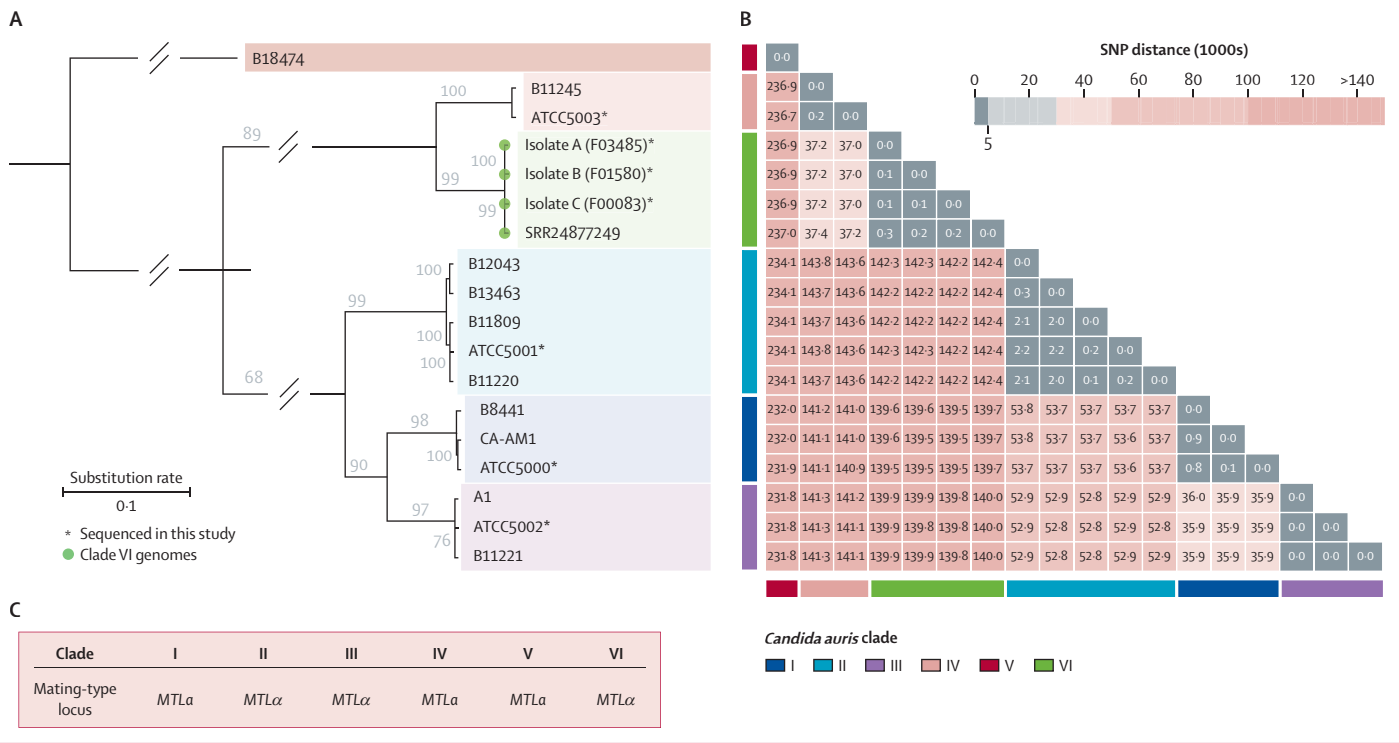


Figure 1: Genome clustering and mating-type alleles of the investigated Singapore *Candida auris* isolates
(A) Genetic relationships among representative *Candida auris* genomes from five known clades and the newly identified sixth clade. Six distinct genetic clusters were recognised, each consisting of genomes from the same clade. Genetic relationships were inferred using maximum-likelihood phylogenetic analysis, with 1000 bootstrap replicates (bootstrap values are shown on each branch). The scale bar represents the mean number of nucleotide substitutions per site. (B) SNP difference among representative *C. auris* genomes. The numbers within each box indicate the number of SNP differences observed between genomes in distinct clades (in 1000s). (C) Summary of mating-type predictions for each clade, determined based on the *MTLa* and *MTLα* alleles present at the mating-type locus. SNP=single nucleotide polymorphism.

SNP distance thresholds based on a subset of genomes with previously reported clade assignment information. The learned threshold could then be used to infer whether any given pair of genomes is from the same clade. A genome relatedness graph was constructed to automatically detect a new clade. In the graph, edges link genomes from the same clade, and genome clusters (or connected components) represent existing and potential new clades. Among 3651 publicly available whole-genome sequences, 1132 (31.0%) had previously reported clade information (appendix 1 p 2). We grouped all publicly available genomes and the three investigated genomes into 19 quarterly datasets based on their release dates (January, 2019, to June, 2023). At each timepoint, genome clusters were identified based on the SNP distance threshold learned by the model. A dendrogram was generated based on whole-genome SNP distance values among 3654 isolates using hierarchical clustering with single-linkage. Nodes were collapsed using iTOL version 6 based on the SNP distance threshold of 21 274 (d_{high} , upper bound of the 95% highest posterior density interval), which was learned from the most recent timepoint (quarter 3 of 2023). Detailed methods regarding the algorithm and model evaluation are in appendix 2 (pp 10–16).

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Details of the three investigated isolates (A, B, and C) are summarised in table 1. All three isolates were identified to be *C. auris* (appendix 2 pp 17–18). The colony and microscopy morphology of these isolates were similar to that observed in clades I–IV and are detailed in appendix 2 (p 19). No phenotypic resistance to antifungals was found among the investigated isolates (table 2 and appendix 2 pp 20–21). No known antifungal resistance-conferring variants were detected (appendix 2 pp 21–22), concordant with the phenotypic test results. All three isolates were found to be non-aggregating. Further results and discussions of the relevant epidemiological investigations, antifungal resistance testing, antifungal resistance genes, and aggregation are in appendix 2 (pp 20–24).

De novo hybrid genome assembly of the *C. auris* genomes was performed for each of the investigated isolates. The total assembly size was 12.4–12.5 Mbp, and each consisted of eight to nine contigs, corresponding to near-complete genomes. Quality statistics are in appendix 1 (p 1). The final dataset for the genome relatedness analysis included 3654 whole-genome sequences after quality control checks, and more than 6.6 million genome pairs were analysed (appendix 1 p 2). Five of the genetic clusters corresponded to the known clades (I–V; appendix 1 p 2). Isolates A, B, and C formed a separate cluster, which we refer to henceforth as clade VI (appendix 1 p 2). One additional *C. auris* genome

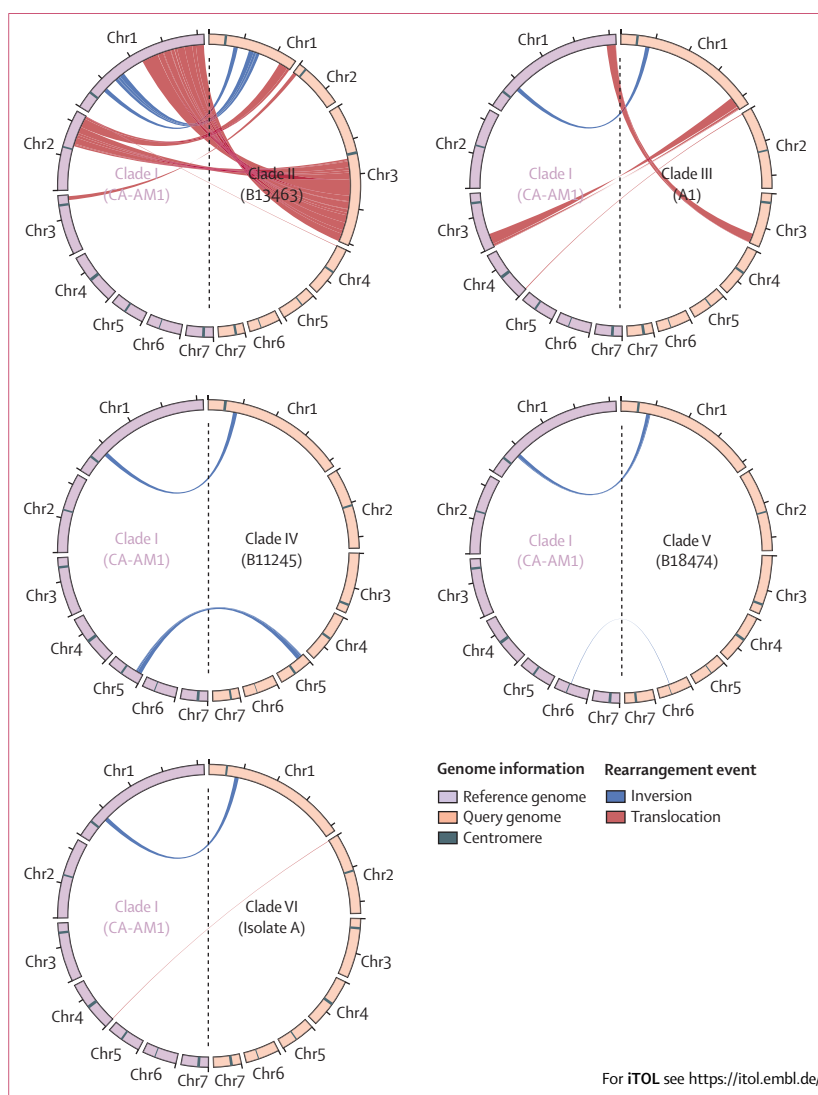


Figure 2: Chromosomal rearrangements in *Candida auris*

Chord diagrams showing translocations and inversions in representative genomes for clades II, III, IV, V, and VI, using CA-AM1 (clade I) as a reference. Only contigs and rearrangement events larger than 10 000 bp are shown. Representative isolate genomes used in this figure were: clade I, CA-AM1 (GCA_014673535); clade II, B13463 (GCA_016495665); clade III, A1 (GCA_014217455); clade IV, B11245 (GCA_008275145); clade V, B18474 (SRR13269545); and clade VI, isolate A (F3485). Chr=chromosome.

(SRR24877249), independently submitted from Bangladesh, also clustered with isolates A, B, and C.

Phylogenetic analysis showed that clade VI was genetically distinct from the five known clades (figure 1A). Clade VI genomes were most closely related to clade IV representatives but differed from all five known clades by at least 37 000 SNPs (range 37 000–236 900; figure 1B). A second GATK-based pipeline similarly suggested that clade VI was different from all five known clades by at least 38 100 SNPs (range 38 100–235 900). Details of SNP distances calculated by the two different pipelines are in appendix 2 (pp 24–25). Additional tables summarising SNP differences between clades are in appendix 2 (pp 26–27). Within the clade VI cluster,

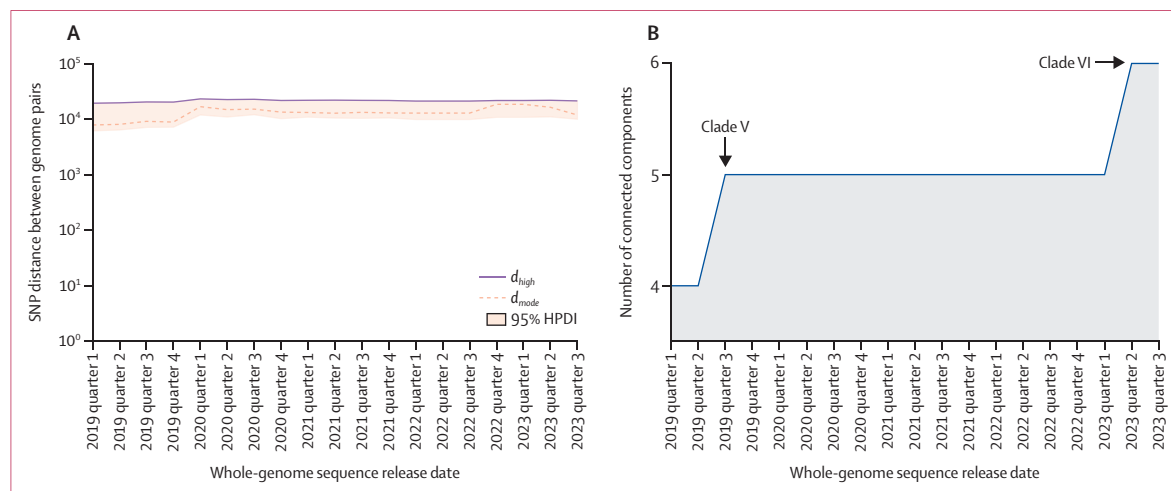


Figure 3: Detection of potentially new *Candida auris* clades using a Bayesian analysis framework

(A) SNP distance thresholds learned across timepoints using Bayesian logistic regression. The pink dashed line represents the mode of the SNP distance threshold (d), and the shaded region represents the 95% HPDI bounded by d_{low} and d_{high} . When the SNP distance is greater than d_{high} , the input genome pair is predicted to be from different clades. *C. auris* whole-genome sequencing data released within the same period were added incrementally into the training dataset. (B) The number of connected components detected over time. The emergence of clade V and clade VI was detected by the increase in the number of connected components (black arrows). HPDI=high posterior density interval. SNP=single nucleotide polymorphism.

the three Singaporean isolates were distinct in at least 68 core genome SNPs (appendix 2 p 27). Similarly, *in silico* STR analysis showed that the clade VI isolates formed a distinct

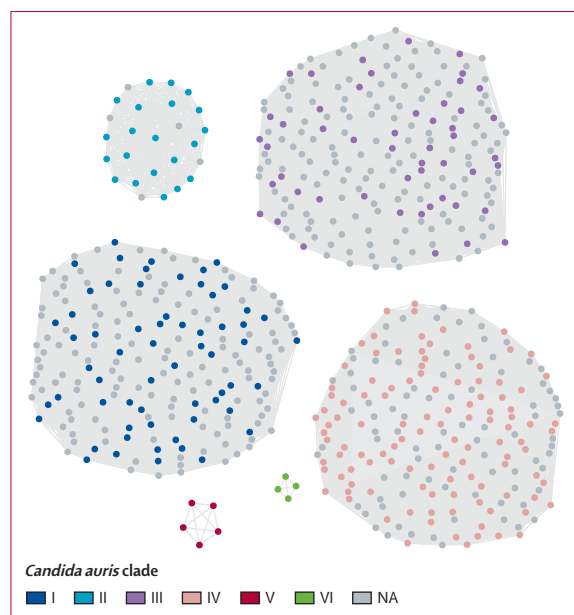


Figure 4: Relationships between *Candida auris* genomes as determined by the Bayesian logistic regression model

Each node in the graph represents a *C. auris* genome, and node colour denotes clades. Each colour represents genomes with known clade information. Grey nodes represent genomes without clade information in the literature. For ease of visualisation, we subsampled the genomes in the connected components that contained more than 200 genomes. An edge links two nodes when the SNP distance is less than the upper bound of the 95% high posterior density interval, indicating that the genomes are predicted to belong to the same clade. Six connected components were observed across 3654 genomes. NA=not available. SNP=single nucleotide polymorphism.

sixth cluster based on STR profiles (appendix 2 p 28). The results suggest that the divergence time of clade VI from the most closely related clade IV was 4831 BCE (95% high posterior density interval [HPDI] 4519–5143), and the estimated mean time to the most recent common ancestor for clade VI was 35.6 years ago (95% HPDI 29.6–40.2). Details of the divergence time estimation are in appendix 2 (p 29).

We further investigated whether the clade VI isolates have clade-specific chromosomal arrangement features. Based on whole-genome alignments, clade VI has an inversion in the long arm of chromosome 1, which is also observed in the representative genomes from clades II, III, IV, and V. We also identified a unique translocation event between chromosomes 2 and 4 in clade VI. Compared with their closest relatives in clade IV, an inversion in chromosome 5 was not observed in the clade VI genomes (figure 2 and appendix 2 p 30). In addition, clade VI (*MTLa*) and clade IV (*MTLa*) were of opposite mating types (figure 1C and appendix 2 p 31).

We then examined whether the investigated isolates were recombinants or hybrids of the known clades (I–V). Based on ADMIXTURE analysis, the cross-validation errors were lowest when $k=7$. At $k=7$, clade VI genomes were distinct and homogeneous in five replicates, suggesting that these genomes were not hybrids or recombinants of the known clades. Details of ADMIXTURE results are in appendix 2 (p 32).

Given the marked difference among genomes of *C. auris* from different clades, we exploited the SNP distances between the genomes to develop a system to monitor and alert for the emergence of potentially new clades. Using publicly available WGS data for 3651 isolates and the three representative genomes from clade VI, with a final dataset of

more than 6.6 million genome pairs, we simulated the detection of clade V and clade VI at two timepoints, when the WGS data became available (figure 3). Although the amount of WGS data increased from 554 genomes in quarter 1 of 2019 to 3654 genomes in quarter 3 of 2023, the SNP distance thresholds for clade differentiation inferred by the Bayesian logistic regression model were consistently between 19 000 and 23 000 SNPs across the timepoints (figure 3A). The model detected the emergence of clades V (quarter 3 of 2019) and VI (quarter 2 of 2023) as it monitored the SNP distance among the genomes over time (figure 3B). As an illustrative example, in quarter 3 of 2023, there were six connected components and each component consisted of genomes from the same clade (figure 4). A similar topology representing the six clades was observed in the dendrogram constructed based on whole-genome SNP distances (figure 5). Detailed results on the automatic detection of new clades are in appendix 2 (p 33).

Discussion

We report the discovery of a proposed sixth major clade of *C. auris*, which includes three epidemiologically unlinked isolates detected in Singapore, and one isolate reported from Bangladesh (SRR24877249) for which WGS data are publicly available. We propose this new clade based on extensive genomic analysis showing the clustering of these four isolates, and great genetic distance from the five known clades. We obtained high-quality hybrid genome assemblies of the Singaporean clade VI isolates and analysed these genomes together with 3651 publicly available *C. auris* whole-genome sequences. The use of WGS and hybrid assembly allowed us to detect and characterise isolates from a potentially novel clade, which is not possible with most conventional clade typing methods (appendix 2 p 34).

Although the samples that yielded these isolates were collected in Singapore and Bangladesh, clade VI isolates were genetically most closely related to clade IV (South American) representatives. The genomic distance observed between clades IV and VI was greater than that observed between clades I and III (around 35 900 SNPs). The independent detection of clade VI isolates from Singapore and Bangladesh suggests that clade VI strains might have been circulating undetected in southeast Asia and south Asia. Given the divergence time estimates from clade IV (4831 BCE; appendix 2 p 34), we speculate that there might exist an endemic zone or a special ecological habitat for clade VI *C. auris* in the region spanning south and southeast Asia within the Indomalaya biogeographic realm (which we henceforth refer to as the Indomalayan zone).

Clade VI isolates were distinct from their closest relatives in clade IV based on phenotypic investigations and genomic analysis. First, the susceptibility of clade VI isolates to antifungals was unlike that of clade IV isolates, approximately half of which are resistant to fluconazole.^{12,13,25,26} Second, clade VI isolates lack an inversion observed in clade IV. Third, clade IV (*MTLa*) and clade VI (*MTLa*) isolates had opposite mating-type alleles. As isolates of opposite

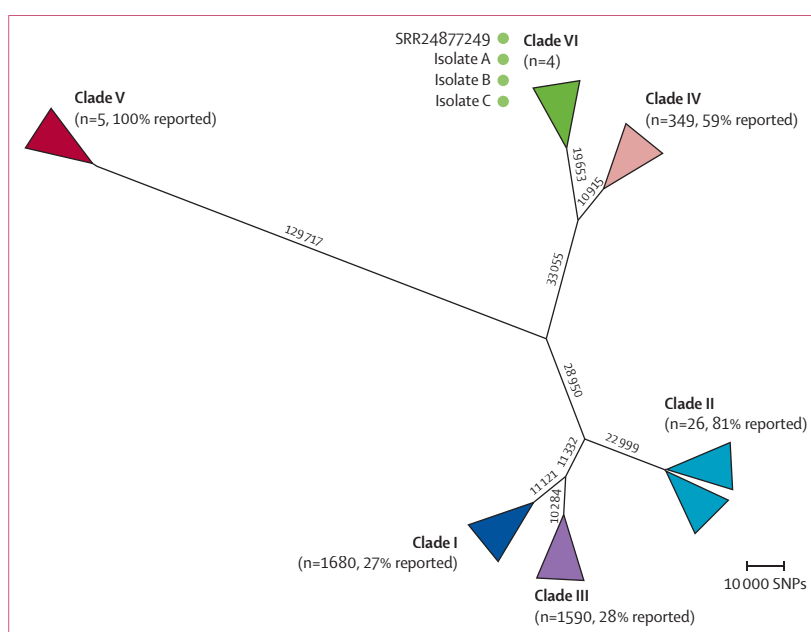


Figure 5: A dendrogram generated based on whole-genome SNP distance among 3654 *Candida auris* genomes Nodes were collapsed using the distance threshold of 21 274 (the upper bound of the 95% high posterior density interval), which was learned from the most recent timepoint (quarter 3 of 2023) through the Bayesian logistic regression model. The number of genomes in each branch, and the percentage of genomes within each branch with previously reported clade assignment information are shown in parentheses. Numbers on each branch indicate genomic distances (SNPs). SNP=single nucleotide polymorphism.

mating types are yet to be found within the same clade, taken together with other findings presented in this study, the clade VI isolates are likely to be representatives of a new major clade rather than a sublineage of an existing clade. Further discussion on mating type can be found in appendix 2 (p 34).

Unlike the five known *C. auris* clades that are named after their geographical locations, the clade VI isolates detected in Singapore have no known specific geographical associations yet. We postulate that isolate C was related to Bangladesh geographically, as patient C was transferred from Bangladesh to Singapore. The independent detection of another clade VI isolate in Bangladesh (SRR24877249, collected in 2021) suggests that clade VI strains might be circulating undetected in Bangladesh, with the first documented clade VI isolate being detected in Singapore (isolate C). Patient B was known to have epidemiological links to Indonesia and Singapore, but not to Bangladesh or south Asia. Patient A was not known to have left Singapore in the preceding 2 years and has no known epidemiological link to south Asia. Given the small sample size, the identification of more clade VI isolates will be necessary for the characterisation of this new clade, and to pinpoint a geographical origin for this emergent clade.

Due to the paucity of *C. auris* data in the region, it is unclear whether clade VI is widespread or expanding from its presumed endemic Indomalayan zone and its potential for causing invasive infections and outbreaks remains unknown.^{2,7,11–13,26–29} Because the source, risk factors for

acquisition, specific geographical associations, virulence, and outbreak potential of clade VI *C. auris* are yet to be established, it is imperative to ensure its early detection and containment for the purpose of patient safety. The implementation of an active surveillance programme for *C. auris* in Singapore General Hospital led to the discovery of the colonisation of the index case, patient A. Since 2020, Singaporean authorities have put in place active surveillance programmes and enhanced infection control measures in acute care hospitals to proactively control the dissemination of *C. auris*. Active surveillance strategies and priorities might need to be reassessed by individual health-care institutions, to balance possible emergent threat containment with health-care resource use. Further discussion on the detection and containment of *C. auris* in Singapore is in appendix 2 (pp 34–35).

Although laboratories are increasingly able to obtain microbial genome sequences, ongoing, large-scale genomic analysis to provide relevant public health alerts continues to be challenging. The previously submitted WGS data of isolate C from Singapore (SRR10102336),¹⁵ and the independent upload from Bangladesh (SRR24877249) were neither identified nor reported as representatives of a potential new clade at the time of writing. The presence of potentially novel, but unrecognised genomic data present in the NCBI database shows a gap in novel pathogen or genotype detection, despite the availability and accessibility of genomic surveillance data. As such, we aimed to augment our ability to detect outlier genomes, such as the clade VI genomes reported in this study, by demonstrating a proof-of-concept machine learning approach that can automatically identify outlier genomes (detection of clades V and VI) to enhance genomic surveillance. Using quality-controlled WGS data files as input, the model outputs clade prediction based on SNP distance. Possible outliers are automatically flagged for further manual investigation, as these genomes might represent evolutionarily divergent novel clades.

This study has several limitations. Due to the retrospective nature of this study, complete exposure history could not be traced. The sample size of clade VI isolates was small, and, therefore, might not be representative of the entire clade. Antifungal susceptibility testing was not performed with a reference broth dilution method but was performed with Sensititre YeastOne plates (appendix 2 pp 20–21). Stringent quality control of data discarded 824 (18.4%) of 4475 of publicly available genome data, including isolate C's sequence (SRR10102336) reported from Singapore in 2019.¹⁵ Furthermore, the use of a single reference genome to identify SNPs might introduce inherent biases. These limitations suggest that when applied in practice, our mathematical modelling approach to detect new clades will face additional challenges. The framework presented in this study therefore represents a proof-of-concept that requires further evaluations, fine-tuning, and user monitoring. Further discussion on methods used for *C. auris* clade differentiation and the results and limitations of this machine learning approach is in appendix 2 (p 35).

Despite its limitations, the approach used in this study allowed us to leverage all available *C. auris* data at a given timepoint to arrive at a decision threshold for classifying genomes into clades, which is unlike the current approach of phylogenetic analysis in which a relatively small number of genomes need to be visualised in a tree structure. With a dynamic and data-driven threshold in place, algorithmic decisions can be made to flag outlier genomes automatically. When incorporated into public microbial genome data repositories, public health laboratories, or large clinical laboratories that generate or have ready access to WGS data, the users can be alerted to unusual outlier genomes as soon as the genome data become available. This capability to generate early alerts is particularly important for regional hubs such as Singapore, where the confluence of a large volume of international visitors positions the city as an ideal sentinel site for identifying emerging public health threats.

Worsening spread of *C. auris* has been reported in the past decade in Europe and in the USA.^{30–32} It is evident that *C. auris* continues to be a threat to public health, notwithstanding the resources available for surveillance and containment in high-income countries. Given the resource limitation in parts of the Indomalayan zone, innovative solutions are necessary to augment regional microbial genomic surveillance. Cost-effective integration of existing routine infectious disease diagnostics with genomic surveillance at sentinel sites,^{33–35} together with effective sequence data analysis, are essential components for pandemic preparedness. Our study provides evidence for the discovery of the sixth major *C. auris* clade in Singapore and underscores the need to enhance genomic surveillance in strategic sentinel sites.

Contributors

CS and KKKK conceptualised the study and CS, KKKK, and CKMT prepared the draft manuscript. CS, KKKK, CKMT, KML, and MGT performed the literature search. KML, CKMT, and NN contributed to the conceptualisation and design of the study. MGT, ALT, and YET contributed isolates for analyses and provided expert advice on clinical mycology. CS, KKKK, KML, PB, SSG, PR, BS, and JJKC contributed to the data acquisition and analysis process. KSC and LLEO contributed samples for analysis and provided expert advice on clinical molecular microbiology. LCL, MLL, KYT, BBSI, MKA, JXYS, YY, and IV provided epidemiological data and expert infection control and prevention advice. BPZC and LW provided specialist infectious disease inputs and clinical data. JFM provided expert advice and supported short tandem repeat and phylogenetic timing analysis. BS performed short tandem repeat and phylogenetic timing analysis. BHT and THK critically evaluated the clinical, microbiological, and epidemiological data and analysis. NN critically evaluated the genomic data analysis. All authors contributed to the manuscript, and edited and reviewed the final manuscript. CS and KKKK accessed and verified the raw data. CS and KKKK contributed equally to this work. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

BHT declares that he serves on the advisory boards for Pfizer (for respiratory syncytial virus vaccine) and MSD (for letermovir). ALT declares that she serves on the Chapter of Pathologists, Singapore. She also received an honorarium (to her institution) for a lecture delivered to the Singapore Association of Pharmaceutical Industries. All other authors declare no competing interests.

Data sharing

The code used to perform whole-genome sequence analyses and train machine learning models in this study is available at https://github.com/CSB5/Candida_auris_CladeVI. Reads and genome assemblies from this study have been deposited in the National Center for Biotechnology Information Sequence Read Archive database under BioProject accession number PRJNA1000034.

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References

- Morales-López SE, Parra-Giraldo CM, Ceballos-Garzón A, et al. Invasive infections with multidrug-resistant yeast *Candida auris*, Colombia. *Emerg Infect Dis* 2017; **23**: 162–64.
- Ruiz-Gaitán A, Moret AM, Tasiatis-Pitarch M, et al. An outbreak due to *Candida auris* with prolonged colonisation and candidaemia in a tertiary care European hospital. *Mycoses* 2018; **61**: 498–505.
- Adams E, Quinn M, Tsay S, et al. *Candida auris* in healthcare facilities, New York, USA, 2013–2017. *Emerg Infect Dis* 2018; **24**: 1816–24.
- Eyre DW, Sheppard AE, Madder H, et al. A *Candida auris* outbreak and its control in an intensive care setting. *N Engl J Med* 2018; **379**: 1322–31.
- Theodoropoulos NM, Bolstorff B, Bozorgzadeh A, et al. *Candida auris* outbreak involving liver transplant recipients in a surgical intensive care unit. *Am J Transplant* 2020; **20**: 3673–79.
- Calvo B, Melo ASA, Perozo-Mena A, et al. First report of *Candida auris* in America: clinical and microbiological aspects of 18 episodes of candidemia. *J Infect* 2016; **73**: 369–74.
- Schelenz S, Hagen F, Rhodes JL, et al. First hospital outbreak of the globally emerging *Candida auris* in a European hospital. *Antimicrob Resist Infect Control* 2016; **5**: 35.
- Welsh RM, Bentz ML, Shams A, et al. Survival, persistence, and isolation of the emerging multidrug-resistant pathogenic yeast *Candida auris* on a plastic health care surface. *J Clin Microbiol* 2017; **55**: 2996–3005.
- Fisher MC, Denning DW. The WHO fungal priority pathogens list as a game-changer. *Nat Rev Microbiol* 2023; **21**: 211–12.
- Centers for Disease Control and Prevention, National Center for Emerging Zoonotic and Infectious Diseases, Division of Healthcare Quality Promotion. Antibiotic Resistance Coordination and Strategy Unit. Antibiotic resistance threats in the United States. 2019. <https://doi.org/10.15620/cdc:82532> (accessed June 4, 2024).
- Spruijtenburg B, Badali H, Abastabar M, et al. Confirmation of fifth *Candida auris* clade by whole genome sequencing. *Emerg Microbes Infect* 2022; **11**: 2405–11.
- Lockhart SR, Etienne KA, Vallabhaneni S, et al. Simultaneous emergence of multidrug-resistant *Candida auris* on 3 continents confirmed by whole-genome sequencing and epidemiological analyses. *Clin Infect Dis* 2017; **64**: 134–40.
- Chow NA, de Groot T, Badali H, Abastabar M, Chiller TM, Meis JF. Potential fifth clade of *Candida auris*, Iran, 2018. *Emerg Infect Dis* 2019; **25**: 1780–81.
- Iguchi S, Itakura Y, Yoshida A, et al. *Candida auris*: a pathogen difficult to identify, treat, and eradicate and its characteristics in Japanese strains. *J Infect Chemother* 2019; **25**: 743–49.
- Tan YE, Teo JQ-M, Rahman NBA, et al. *Candida auris* in Singapore: genomic epidemiology, antifungal drug resistance, and identification using the updated 8.01 VITEK®2 system. *Int J Antimicrob Agents* 2019; **54**: 709–15.
- Hernando-Ortiz A, Mateo E, Perez-Rodriguez A, de Groot PWJ, Quindós G, Eraso E. Virulence of *Candida auris* from different clinical origins in *Caenorhabditis elegans* and *Galleria mellonella* host models. *Virulence* 2021; **12**: 1063–75.
- Treangen TJ, Ondov BD, Koren S, Phillippy AM. The Harvest suite for rapid core-genome alignment and visualization of thousands of intraspecific microbial genomes. *Genome Biol* 2014; **15**: 524.
- Li X, Muñoz JF, Gade L, et al. Comparing genomic variant identification protocols for *Candida auris*. *Microb Genom* 2023; **9**: mgen000979.
- Muñoz JF, Welsh RM, Shea T, et al. Clade-specific chromosomal rearrangements and loss of subtelomeric adhesins in *Candida auris*. *Genetics* 2021; **218**: iyab029.
- Muñoz JF, Gade L, Chow NA, et al. Genomic insights into multidrug-resistance, mating and virulence in *Candida auris* and related emerging species. *Nat Commun* 2018; **9**: 5346.
- de Groot T, Spruijtenburg B, Parnell LA, Chow NA, Meis JF. Optimization and validation of *Candida auris* short tandem repeat analysis. *Microbiol Spectr* 2022; **10**: e0264522.
- Spruijtenburg B, Bombassaro A, Meijer EFJ, et al. *Sporothrix brasiliensis* genotyping reveals numerous independent zoonotic introductions in Brazil. *J Infect* 2023; **86**: 610–13.
- Alexander DH, Novembre J, Lange K. Fast model-based estimation of ancestry in unrelated individuals. *Genome Res* 2009; **19**: 1655–64.
- Wang Y, Xu J. Population genomic analyses reveal evidence for limited recombination in the superbug *Candida auris* in nature. *Comput Struct Biotechnol J* 2022; **20**: 3030–40.
- Chowdhary A, Prakash A, Sharma C, et al. A multicentre study of antifungal susceptibility patterns among 350 *Candida auris* isolates (2009–17) in India: role of the ERG11 and FKS1 genes in azole and echinocandin resistance. *J Antimicrob Chemother* 2018; **73**: 891–99.
- Chow NA, Muñoz JF, Gade L, et al. Tracing the evolutionary history and global expansion of *Candida auris* using population genomic analyses. *MBio* 2020; **11**: e03364-19.
- Chow NA, Gade L, Tsay SV, et al. Multiple introductions and subsequent transmission of multidrug-resistant *Candida auris* in the USA: a molecular epidemiological survey. *Lancet Infect Dis* 2018; **18**: 1377–84.
- Kwon YJ, Shin JH, Byun SA, et al. *Candida auris* clinical isolates from South Korea: identification, antifungal susceptibility, and genotyping. *J Clin Microbiol* 2019; **57**: e01624-18.
- Ben Abid F, Salah H, Sundararaju S, et al. Molecular characterization of *Candida auris* outbreak isolates in Qatar from patients with COVID-19 reveals the emergence of isolates resistant to three classes of antifungal drugs. *Clin Microbiol Infect* 2023; **29**: 1083.e1–7.
- Lyman M, Forsberg K, Sexton DJ, et al. Worsening spread of *Candida auris* in the United States, 2019 to 2021. *Ann Intern Med* 2023; **176**: 489–95.
- Kilburn S, Innes G, Quinn M, et al. Antifungal resistance trends of *Candida auris* clinical isolates in New York and New Jersey from 2016 to 2020. *Antimicrob Agents Chemother* 2022; **66**: e0224221.
- Kohlenberg A, Monnet DL, Plachouras D. Increasing number of cases and outbreaks caused by *Candida auris* in the EU/EEA, 2020 to 2021. *Euro Surveill* 2022; **27**: 2200846.
- Ko KKK, Chu JJK, Lim KM, et al. Clonal serotype 1c multidrug-resistant *Shigella flexneri* detected in multiple institutions by sentinel-site sequencing. *Front Med (Lausanne)* 2022; **9**: 964640.
- Ko KKK, Chng KR, Nagarajan N. Metagenomics-enabled microbial surveillance. *Nat Microbiol* 2022; **7**: 486–96.
- Yingtaeweessittikul H, Ko K, Abdul Rahman N, Tan SYL, Nagarajan N, Suphavitai C. CalmBelt: rapid SARS-CoV-2 genome characterization for outbreak tracking. *Front Med (Lausanne)* 2021; **8**: 790662.

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